NOT MEASUREMENT SENSITIVE

MIL-A-46153B
AMENDMENT 2
18 January 1990
SUPERSEDING
AMENDMENT 1
17 March 1981

MILITARY SPECIFICATION

ANTIFREEZE, ETHYLENE GLYCOL, INHIBITED, HEAVY DUTY, SINGLE PACKAGE

This amendment forms a part of Military Specification MIL-A-46153B, dated 31 July 1979, and is approved for use by all Departments and Agencies of the Department of Defense.

PAGE 1

* 1.2, under SPECIFICATIONS, FEDERAL, delete O-I-490 and PPP-C-300 in their entirety.

PAGE 2

* 1.2, under SPECIFICATIONS, MILITARY, delete MIL-STD-1188 in its entirety.

* 2.2, under AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM), delete D 2111 in its entirety and add:

"D 1122 - Specific Gravity of Engine Coolants by the Hydrometer.
D 1238 - Flow Rates of Thermoplastics by Extrusion Plastometer."

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* After 4.6.1, add:

"4.7 Inspection of packaging.

"4.7.1 Quality conformance inspection of packing.

"4.7.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

"4.7.1.2 Examination. Each unit of product shall be examined for the following defects.

101. Preservation not as specified for level A (see 5.1).
102. Packing not as specified for level A, B, or C (see 5.2).
103. Marking incorrect, illegible or missing (see 5.3).
104. Radiator tags not as specified (see 5.4).
105. Palletization not as specified when required (see 5.5)."

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* 5.1 thru 5.5, delete in their entirety and substitute:

"5.1 Preservation. Preservation shall be level A.

"5.1.1 Level A.

"5.1.1.1 One quart quantity. One quart antifreeze shall be unit packed in a container conforming to PPP-C-569, type I, class A, as specified in its appendix thereto or in a one quart capacity polyethylene plastic bottle conforming to MIL-B-26701. If the bottle conforming to MIL-B-26701 is being used to package composition A, the following requirements shall be mandatory. An additive shall be used in the polyethylene resin which will yield a bottle of MIL-B-26701 no portion of which will transmit more than 1 percent of ultraviolet light at any wavelength in the range of 310 to 325 millimicrons when tested using a calibrated spectrophotometer with air as a reference (see 6.3). If composition B, the ultraviolet additive is not required.

"5.1.1.2 One gallon quantity. One U.S. gallon of antifreeze shall be unit packed as specified for the one quart quantity except the plastic container and box specified in PPP-C-569 shall be type 1, class B and the bottle specified in MIL-B-26701 (with the ultraviolet exclusion additive) shall conform to the one gallon capacity size. A leak-proof inner seal which shall prevent leakage of container contents without benefit of the screw-cap closure when the container is inverted shall be furnished beneath the screw-cap of each container.

"5.1.1.3 Five gallon quantity. Five U.S. gallons of antifreeze shall be unit packed as specified in PPP-C-1337 in a type II, class 1 metal container fitted with a polyethylene insert, or MIL-D-43703 size I container. A leak-proof, either inner seal or pilfer-proof locking seal shall be furnished with the screw-cap closure of each container. The filled container shall be tightly closed to prevent leakage of contents.

"5.1.1.4 Fifty-five gallon quantity. Fifty-five U.S. gallons of antifreeze shall be unit packed as specified in PPP-C-1337 in a type II, class 4 metal container fitted with a polyethylene insert or a nominal 55 gallon capacity polyethylene drum conforming to PPP-D-1860. The PPP-D-1860 polyethylene shall have a specific gravity of 0.941 to 0.965 and a melt index of 0.20 to 1.2 gm per 10 minutes using ASTM D 1238 procedure A, condition E. The filled container shall be tightly closed with rigid plastic fittings to prevent leakage of contents.

"5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

"5.2.1 Level A.

"5.2.1.1 One quart and one gallon quantities. Twelve one quart or six one gallon bottles of antifreeze unit preserved as specified in 5.1.1.1 and 5.1.1.2, respectively, shall be packed within a close fitting box conforming to PPP-B-601, type overseas. Interlocking partitions, fabricated from fiberboard shall be utilized to separate bottles as required.

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"5.2.1.1 Five gallon and fifty-five gallon quantities. The five and fifty-five gallon quantities shall require no further packing than that specified in 5.1.1.3 and 5.1.1.4, respectively.

"5.2.2 Level B.

"5.2.2.1 One quart and one gallon quantities. Packing shall be as specified in 5.2.1.1, except exterior box shall conform to PPP-B-636, class weather resistant, variety single wall, grade optional.

"5.2.2.2 Five-gallon and fifty-five gallon quantities. The five and fifty-five gallon quantities shall require no further packing than that specified in 5.1.1.3 and 5.1.1.4, respectively.

"5.2.3 Level C.

"5.2.3.1 One quart and one gallon quantities. Packing shall be as specified in 5.2.1.1, except the exterior box shall be in accordance with PPP-B-636, class domestic, variety single wall, grade optional.

"5.2.3.2 Five-gallon and fifty-five gallon quantities. The five and fifty-five gallon quantities shall require no further packing than that specified in 5.1.1.3 and 5.1.1.4, respectively.

"5.3 Marking. Marking shall be in accordance with MIL-STD-129. In addition, each container shall be marked with the information as shown in figure 1.

"5.4 Protection table and radiator tags. Unless otherwise specified (see 6.2), protection tables and radiator tags for the antifreeze shall be provided on, or with, containers of antifreeze in the following quantities. Not less than 4 tags or labels shall be furnished with each 24 one quart containers or 6 one gallon containers, 3 tags or labels with each 5 gallon container, and 20 tags or labels with each 55 gallon container. Pressure sensitive labels may be used in lieu of radiator tags. Radiator tags shall conform to UU-T-81, type B, grade 20R, wire strung size 5 and shall be marked on side 1 and 2 as shown in figure 2. Radiator labels shall conform to MIL-F-16377 except that the adhesive shall be water-insoluble, pressure-sensitive, and protected by release paper backing. These labels shall be 6 cm wide and 15 cm long ±2 mm and shall be marked as shown in figure 3. Tags or labels shall be packed together in a bag in accordance with MIL-P-116, method IC-3. The bag shall be placed in the box or on the drum, as applicable. When placed in a box, the bag shall be placed flat on the top of the contents and beneath the box flaps. When placed on the drum, the bag shall be secured to the drum with tape conforming to PPP-T-60, type IV, class 1, in an area not normally exposed to abrasion.

"5.5 Palletization. When specified (see 6.2), like packs of one quart, one gallon, and five gallon quantities shall be palletized. Level A, B, and C packs shall be palletized in accordance with the applicable requirements including those for opaque shrink film bonding of MIL-STD-147, using the pallet conforming to NN-P-71, type IV, size 2. The alternate five gallon polyethylene container conforming to MIL-D-43703 shall be palletized using the vertical frame supports

in the form of side frames, storage aid 10, and frame supports, storage aid 13

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with the wood cap storage aid 5. A maximum of 24 five gallon drums shall be placed on a pallet."

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* 6.1, delete in its entirety and substitute:

6.1 Intended use. Inhibited ethylene glycol (i.e., undiluted antifreeze) is intended for use in the cooling systems of liquid-cooled internal combustion engines, other than aircraft, for protection against freezing in ambient temperatures as low as -48 °C when diluted to 60 percent by volume with water. It may also be used as a coolant in some types of automatic guns such as water-cooled machine guns. This antifreeze is intended for use in lieu of O-A-548, type 1. This antifreeze is essentially O-A-548, type 1 plus inhibitor conforming to MIL-A-53009.

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* After 6.3, add the following:

"6.3 Level of preservation. Levels of preservation have not been included as the preservation specified is the minimum acceptable for any level of protection."

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Figure 2, side 1: Delete, "REPLACE AFTER ONE YEAR'S USE."

Figure 2, side 2: Delete, "REPLACE AFTER ONE YEAR's USE."

Figure 3: Delete, "REPLACE AFTER ONE YEAR'S USE."

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